Task 2. Focusing through a simulated bench

Code: https://github.com/mariammnatsakanyan09-dot/Task-5-6/tree/main/hw2_task2

Video: hw2_task2/video/task2_demo.mp4

Model: nvidia/nemotron-3-super-120b-a12b:free through the OpenRouter API (free key), the same model in all four roles.

What was built. A simulated bench (laser, DMD with amplitude only, scattering medium, camera with shot and read noise; E_out = |T E_in|^2) and a tool `focus_search` that runs the whole search of the task on it: macropixels, a random start, the intensity in the focus region and in the proximal background (a ring split into regions), the enhancement as focus against background, flipping macropixels and keeping the flips that raise the enhancement. The tool returns the enhancement, re-measured on the final pattern with fresh shots, and the camera images. It is added to the executor of the Homework 1, Task 6 system (orchestrator, scientist, general agent, executor on one LangGraph graph). I typed one request, "*Focus the beam.*", and nothing else: the agent chose the macropixel size, the region sizes and the number of iterations itself, called the tool 4 times (4 successful experiments) in 4 orchestrator decisions, and kept the enhancement the tool returned.

Experiments the agent ran

#	macropixel (N)	focus px	ring gap / width / regions	iterations	flip / avg	enhancement start -> final	vs far field	focus gain	background ratio
1	4 (256)	3	2 / 4 / 4	200	1 / 4	**1.37 -> 29.1 ***	13.5	9.3	0.44
2	8 (64)	3	2 / 4 / 4	200	1 / 4	1.49 -> 22.0	12.6	5.1	0.35
3	16 (16)	3	2 / 4 / 4	200	1 / 4	0.69 -> 6.4	5.1	5.0	0.54
4	8 (64)	3	2 / 4 / 4	800	1 / 4	1.17 -> 24.7	10.0	4.1	0.20

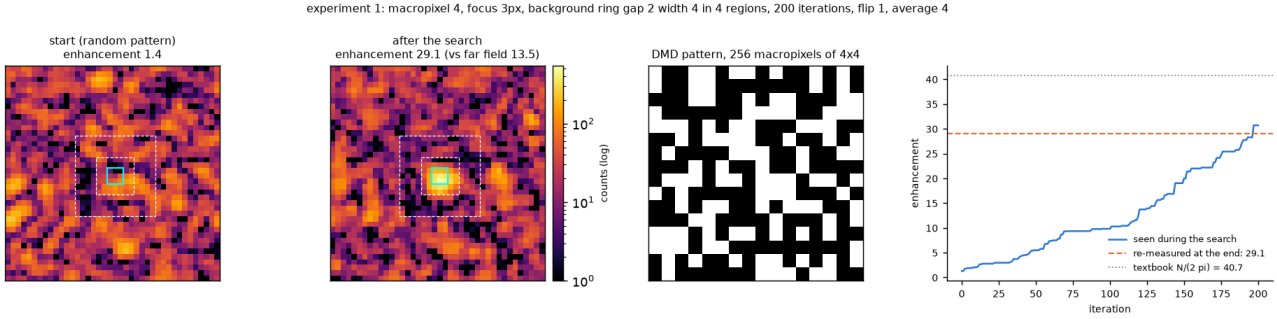
Enhancement = mean focus intensity / mean of the background regions, on the final pattern, re-measured with 50 fresh shots; 'start' is the same number for the random starting pattern. 'vs far field' is the focus intensity divided by the mean intensity of the camera pixels outside the ring (never used by the search). 'focus gain' and 'background ratio' are the mean counts in the focus and in the background ring after the search divided by the same counts before it. * best experiment. Every number is copied from the tool's records (experiments.json).

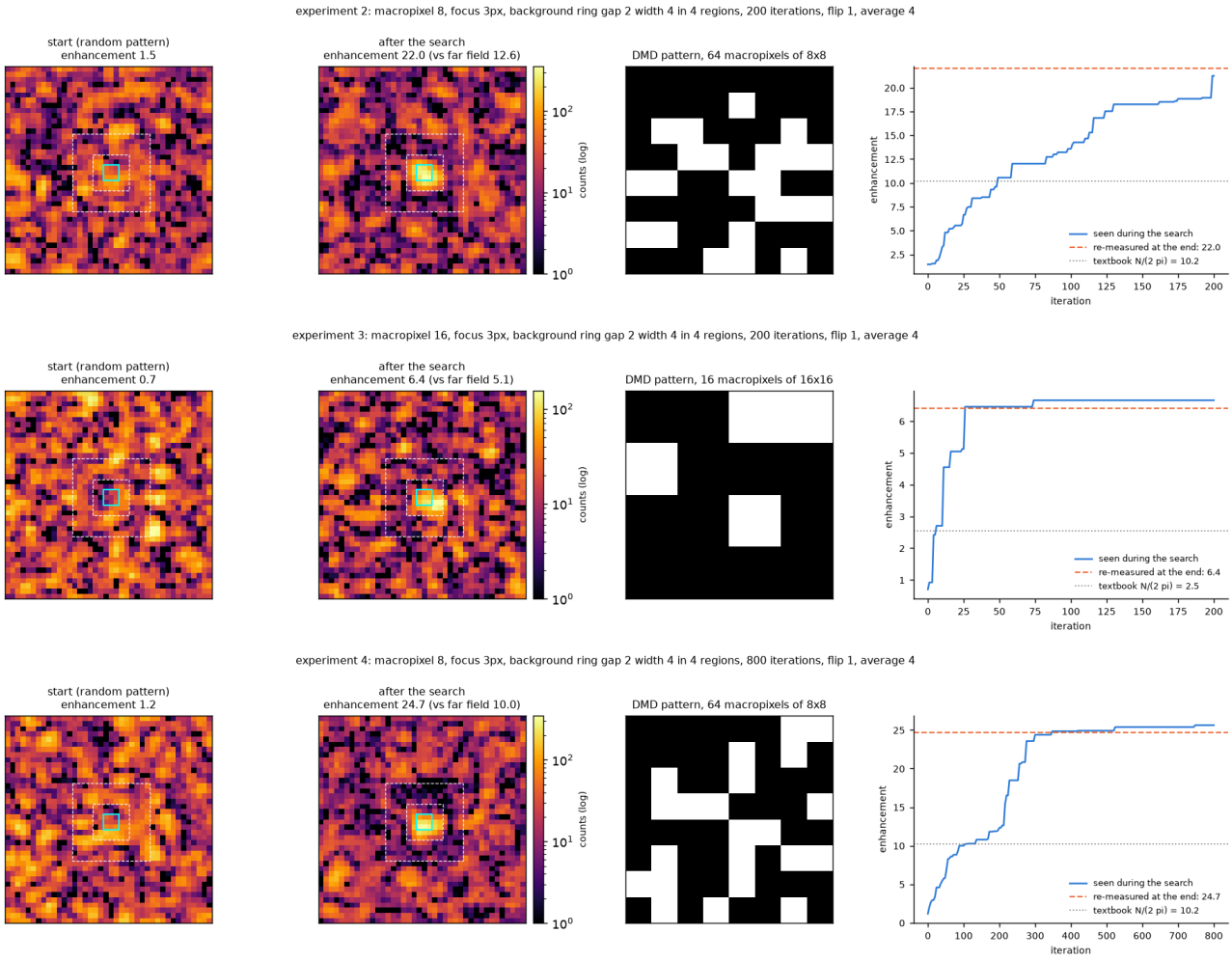
What the agent concluded (final message of the orchestrator)

```
We ran four focus_search experiments varying macropixel_size and iterations while keeping focus 3x3
pixels at (20,20), background gap=2, width=4, 4 sectors, n_flip=1, n_avg=4. Results: macropixel_size=4,
iterations=200 gave enhancement 29.057 (starting 1.37); macropixel_size=8, iterations=200 gave 22.001
(start 1.49); macropixel_size=16, iterations=200 gave 6.409 (start 0.69); macropixel_size=8,
iterations=800 gave 24.703 (start 1.17). The best enhancement observed is 29.057 from experiment 1
(macropixel_size=4, iterations=200). Enhancement falls as macropixel size grows (more control yields
higher enhancement) and rises with more iterations for a fixed macropixel size (compare 8-pixel cases).
Next step: repeat the best setting to check reproducibility and then increase the number of iterations
for macropixel_size=4 to see if enhancement can be further improved.
```

Camera images

Each figure: the camera frame for the random starting pattern, the frame after the search (log scale; cyan = focus region, dashed = background ring), the DMD pattern found, and the enhancement during the search.





Comparison with the focusing result from the lecture

	macropixels N	enhancement	textbook N/(2 pi)	enhancement / (N/(2 pi))
lecture	not given	33	-	-
this run, best (exp. 1), ring background as in the task	256	**29.1**	40.7	0.71
same experiment, against the far field	256	13.5	40.7	0.33

The best enhancement of the agent, 29.1 (experiment 1, 256 macropixels), is somewhat lower than the lecture's 33 (ratio 29.1 / 33 = 0.88).

What the enhancement is made of in the best experiment: the mean focus intensity rose by a factor 9.3 (from 36.1 to 336.1 counts) and the mean background fell by a factor 2.3 (from 26.3 to 11.6 counts). The search keeps a flip when the ratio focus/background rises, so it lowers the proximal background as well as raising the focus. Against the camera area that the search never used (the far field) the same focus is 13.5 times brighter than the background there, which is the enhancement of the focus without the lowered ring.

Enhancement by macropixel size across the experiments: macropixel 16 (16 macropixels): 6.4 (200 iterations); macropixel 8 (64 macropixels): 22.0 (200 iterations), 24.7 (800 iterations); macropixel 4 (256 macropixels): 29.1 (200 iterations).